

MCA- Semester I
Flash Test – CS 404: Discrete Mathematics

Date: 25/08/2025
Time: 9:30 – 10:00am

ANSWER ALL OF THE FOLLOWING

1. Write the names of concepts (in one word) the following denote;
 - a. (a, b, c, d, e) –
 - b. $\{a, b, c, d, e\}$ – Set
 - c. $[a, b, c, d, e]$ –
 - d. $\exists, \forall$ –

2. Put down one common and one uncommon properties of the following

	Common	Uncommon
Partition		
Cover		

3. Give the full form of the following acronyms

- a. wff -
- b. CNF –
- c. CSG
- d. 1-1 –
- e. DAG
- f. Poset –

4. Give an example of the following

- a. Recurrence relation
- b. Recursive function
- c. Homomorphism

5. Differentiate between the following

- a. Characteristic relation and Characteristic function

School of Computer and Systems Sciences
MCA – Semester–I (Monsoon 2026)
CS – 404: Discrete Mathematics

Quiz-I

Date: 17-09-2026

Max Marks: 10

Attempt questions for 10 marks only.

1. Let $A = \{1, 2, 3, 4\}$. Let $R_1 = \{(a, b) \mid a \text{ divides } b\}$, $R_2 = \{(a, b) \mid a \leq b\}$ be relations on A .

a. How are R_1 and the Cartesian product of A related?

_____ [1]

b. Give $R_1 \cap R_2$.

_____ [2]

c. Give two properties of relations common to R_1 and R_2 . Justify your answers.

_____ [3]

2. Let A and B be two arbitrary sets. Let R_3 be a relation from A to B . Give the necessary condition to define R_4 to be compose relations R_3 and R_4 .

_____ [1]

3. Make two remarks about the following, [2]

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4. Let P and R be two arbitrary sets. Let f_1 and f_2 be two function from $P \rightarrow R$.

a. What is the type of f_1 when R is Codomain of f_1 ? _____ [1]

b. What is the type of f_2 when R is Range of f_2 ? _____ [1]

5. Consider the characteristic function C for a set S of 10 elements. [1.5]

a. What is the domain of C ? _____

b. What is the structure of Range of C ? _____